

INDIAN INSTITUTE OF TECHNOLOGY DELHI
INTRODUCTION TO ELECTRICAL ENGINEERING
| AELL101 QUIZ-4 |

NAME:

ENTRY NO:

DATE: 13-05-2025 (9 AM – 9:45 AM)

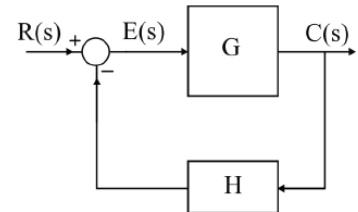
DURATION: 45 min

Total marks: 15

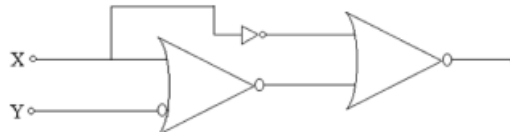
- ✓ Each question carries equal marks. Space for rough work is given on the last page.
- ✓ MCQ (single choice): CIRCLE correct option, otherwise no marks.

1. For the closed-loop system shown, the transfer function $\frac{E(s)}{R(s)}$ is

- A. $\frac{G}{1+GH}$ B. $\frac{GH}{1+GH}$ C. $\frac{1}{1+GH}$ D. $\frac{1}{1+G}$



2. The logic circuit shown below can be minimized to



- A. X' B. X C. XY' D. $X'Y$

3. The Boolean expression $XY + XZ' + YZ$ simplifies to

- A. $YZ + XZ'$ B. $XY + XZ' + Y$ C. $XY + XZ'$ D. $XY + YZ$

4. The output of a logic gate is "1" when all its inputs are at logic "0". The gate is either

- A. NAND or EX-OR B. NOR or EX-OR C. AND or EX-NOR D. NOR or EX-NOR

5. Which of the following is not an analog device?

- A. Thermometer with mercury column B. Light switch
 C. Audio microphone D. Photocopy machine

6. Which of the following statements are TRUE ?

- P: Analog signals are more immune to noise than the digital signals
 Q: Analog signals contain higher resolution information than the digital signals
 R: Digital signals are more immune to noise than the analog signals
 S: Digital signals are higher resolution than the analog signals

- A. P and R B. P and S
 C. Q and R D. Q and S

7. Hysteresis loop shows the relationship between:

- A. Magnetic field strength and magnetic flux
 B. Magnetic field strength and magnetic flux density
 C. Current and magnetic flux density
 D. Voltage and magnetic flux density

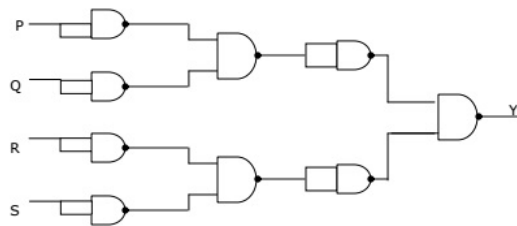
8. Two magnetic circuits are identical in geometry and winding turns, but one has a core made of material X and the other of material Y. The established magnetic flux in both is equal, but the MMF required in circuit X is greater than in circuit Y, what among the following possibilities you can conclude from this information?
- P) Material X has higher permeability than material Y
 Q) Material X has lower permeability than material Y
 R) Current in coil on core X is higher than the coil on core Y
 S) Material X is saturated

- A. P and R
 B. Q and R
 C. P, R and S
 D. Q, R and S

9. 8-bit 2's complement representation of a decimal number is 11111000. The decimal number is _____

Ans: _____

10. For the circuit shown in figure, the simplest Boolean expression for the output Y in terms of inputs P, Q, R and S is _____



Ans: _____

11. Convert 101010 binary number to decimal.

Ans: _____

12. A magnetic circuit has cross-sectional area of 50 cm^2 and mean length of 2 m. The relative permeability of core material is 80. Calculate the reluctance of the magnetic circuit. Permeability of air is $4\pi \times 10^{-7}$

Ans: _____ Ampere-turns per Weber

13. A ring having a cross-sectional area of 500 mm^2 , a circumference of 400 mm and $\phi = 800 \text{ } \mu\text{Wb}$ has a coil of 200 turns wound around it. Relative permeability of ring is 380. Calculate the magnetising current in Ampere.

Ans: _____

14. A transformer has 200 turns on primary side and 20 turns on secondary side. If a 200 V rms AC is applied at the primary side, then what will be the secondary voltage?

Ans: _____

15. A 12 kVA rated, single-phase transformer with transformation ratio $(N_2/N_1) = 0.1$ is fed from 2.5 kV rms supply. It is supplying power to the loads with 0.90 lagging power factor. Determine the primary side current magnitude at full load.

Ans: _____